Supply Chain Traceability for Farm Produce Using QR Code Verification System

Abstract:

In modern agricultural supply chains, the lack of transparency and traceability from farm to consumer creates challenges in verifying the origin, quality, and authenticity of farm produce. This leads to issues such as counterfeit products, exploitation of farmers, and reduced consumer trust. To address these gaps, this research proposes a **QR Code-Based Supply Chain Traceability System** for farm produce. The system assigns a **unique QR code to each batch of produce**, embedding critical information including farmer identity, farm location, cultivation practices, and harvest details. As produce moves through the supply chain—from farmers to distributors, and finally consumers—each transaction is logged and updated in a **centralized blockchain-backed database** to ensure immutability and security.

Stakeholders can scan the QR code at any stage to access real-time data about the produce's journey, ensuring **transparency, authenticity, and accountability**. This not only empowers consumers to make informed choices but also safeguards farmers by providing proof of origin and fair trade compliance. The system leverages QR technology, secure data storage, and traceability protocols to build a **reliable, tamper-proof agricultural supply chain**.

Keywords:

Supply Chain Traceability, QR Code, Farm Produce Authentication, Blockchain, Food Safety, Provenance, Transparency, Secure Data, Agricultural Supply Chain, Consumer Trust.

Problem Statement

Traditional agricultural supply chains often suffer from a lack of transparency and traceability, making it difficult to verify the origin, authenticity, and quality of farm produce. This gap leads to issues such as counterfeit products, contamination risks, exploitation of farmers, and loss of consumer trust. Additionally, farmers often do not receive due credit or fair compensation for their produce as intermediaries dominate the supply chain. Manual documentation processes are error-prone, inefficient, and susceptible to manipulation. Therefore, there is a need for a **secure, automated, and transparent traceability system** that can track farm produce throughout the supply chain, from the farmer to the end consumer, ensuring authenticity, fair trade, and food safety.Working of the System**.**

Step-by-Step Workflow:

1. **Farmer Registration and Produce Tagging**

- Farmers register with the system by submitting identity, farm location, and crop details.
- Upon harvest, each produce batch is tagged with a **unique QR code** that includes farmer ID, crop type, harvest date, and farm location.

2. **Supply Chain Event Logging**

- At each point in the supply chain (e.g., farmers, consumers), the QR code is scanned.
- Every transaction updates the **centralized traceability database** with location, timestamp, and handling details.

3. **QR Code Scanning and Data Access**

- Stakeholders including buyers, and consumers can scan the QR code using a smartphone.

  Scanning retrieves the complete traceability history of the produce, including its origin, transit points, and handling conditions.

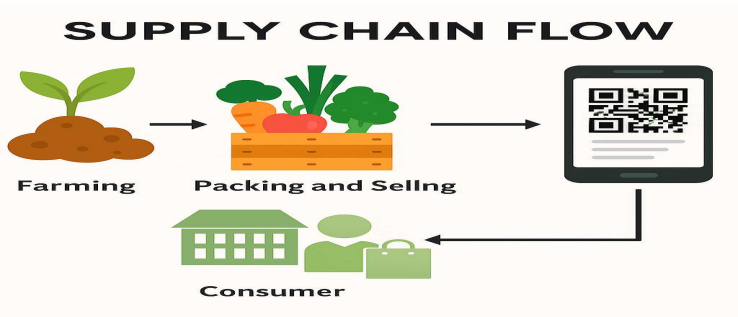
4. **Blockchain Integration for Data Integrity**

- All traceability data is logged on a **blockchain network** to ensure **tamper-proof and immutable records**.
- This guarantees that no party can alter the history of the produce once recorded.

5. **Reports and Analytics**

- The system generates detailed reports for regulatory authorities, supply chain managers, and farmers.
- Visual dashboards provide insights on produce flow, transaction logs, and verification records to ensure compliance

System Architecture Diagram



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